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ITDS 1371

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Module 7 Reflection Journal

This week's module was to learn how to evaluate machine learning models using metrics that provide more insight than accuracy alone.

With the titanic dataset, I trained a Logistic Regression model to predict passenger survival using the features Age, Pclass, Sex, and Fare. After preparing the dataset and handling missing values in the Age column, I split the data into training and testing sets and generated predictions using the trained model. Next, I created a confusion matrix and visualized it using a heatmap. This helped identify the number of true positives(90), true negatives(15), false positives(20), and false negatives(54) made by the model. For task 2, I generated a classification report to evaluate precision, recall, and F1-score for each class. The results were as follows:

	precision	recall	F1-score	support
0	0.82	0.86	0.84	105
1	0.78	0.73	0.76	74
accuracy			0.80	179
Macro avg	0.80	0.79	0.80	179
Weighted avg	0.80	0.80	0.80	179

The model achieved an overall accuracy of approximately 80%. The report showed that the model performed slightly better at identifying passengers who did not survive than those who did survive.

Finally, I used 5-fold cross-validation to obtain a more reliable estimate of model performance. The model achieved an average cross-validation score of 78.68% with a standard deviation of 0.0094, indicating relatively consistent performance across multiple data splits.

In this module, I explored confusion matrices, precision, recall, F1-scores, and cross-validation while working with the Titanic survival dataset. The lab strengthened my understanding of model evaluation techniques and demonstrated why different performance metrics are necessary when assessing machine learning models.

Module 07 Lab - Model Evaluation/Knowledge Check - Diane Lenhoff

1. Describe a real-world scenario where you would care more about a model's Precision than its Recall.

An example would be an email spam filter. Precision is important because we do not want important emails incorrectly classified as spam. A high precision score means that when the model identifies an email as spam, it is usually correct.

2. Describe a real-world scenario where you would care more about a model's Recall than its Precision.

An example would be detecting cancer in medical screenings. Recall is more important because we want to identify as many actual cancer cases as possible. Missing a positive case could have serious consequences. It is better to have some false positives that require additional testing than to miss someone who has cancer.

3. Why is a cross-validation score generally more trustworthy than a score from a single train-test split?

Cross-validation is generally more reliable because it evaluates the model using multiple splits of the dataset instead of just one. This reduces the chance that the results are influenced by a particular split of the data. It also gives us a better understanding of how consistently the model performs on different subsets of data.

Seth Alvarez

Professor Rao

ITAI 1371

October 21, 2026

L07 Reflection Journal: Model Evaluation

In this lab I trained a model on the Titanic dataset and then measured its performance with a confusion matrix, a classification report and cross-validation. I first learned where precision and recall matter more in my computer vision course, and this lab reinforced that idea. I saw the numbers from a single split and from cross-validation side by side. The split gave 80% accuracy, but the five folds scored between, 77.5% and 80.3%, with an average of 78.68% and a standard deviation of 0.0094. The confusion matrix also showed 15 false positives and 20 false negatives, which accuracy alone would fail to surface. This taught me that one split could look better than the real performance, and that I need more than one number to judge a model.

For my Computer Vision final project, I built a baby monitor that uses CV to trigger an alert and help determine the safety status of a baby in their crib. Since my program is a safety system, a trustworthy evaluation matters. This lab makes me want to revisit and check my project, because I haven’t confirmed whether I used a single split or cross-validation, so I should revisit and examine my results again with this lab in mind.

Lab 07 Reflection Journal: Better Model Evaluation

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Course: ITAI1371
Lab: Module 07 Model Evaluation

What I Did in This Lab

In this lab I trained a logistic regression model on the Titanic dataset to predict whether a passenger survived, using age, passenger class, sex, and fare as features. Instead of stopping at accuracy, I used three more detailed ways to evaluate the model: a confusion matrix, a classification report with precision, recall, and F1-score, and 5-fold cross-validation.

What the Results Showed

The model scored about 0.80 accuracy on the test set of 179 passengers, which looks good at first. The classification report showed a more detailed picture. For passengers who died, precision was 0.82 and recall was 0.86. For passengers who survived, precision was 0.78 and recall was 0.73. So the model is better at identifying who died than who survived, and accuracy alone hid that difference. The confusion matrix helped me see exactly where the mistakes were, since I could separate false positives (predicting survival when the person died) from false negatives (predicting death when the person survived).

The cross-validation results were also interesting. The five fold scores ranged from about 0.775 to 0.803, with a mean accuracy of 0.7868 and a standard deviation of 0.0094. My single train-test split gave 0.80, so it was a little higher than the cross-validated average. This told me the single split was slightly lucky, and that the 0.7868 average is a more honest estimate of how the model would do on new passengers. The small standard deviation also showed that the model is fairly stable across different slices of the data.

Challenges I Faced

The biggest challenge was not about the machine learning at all. When I first pasted the starter code into Colab, all of my cells showed a red error mark. The problem was that the code had been pasted onto a single line with no line breaks, so Python read things like "pdfrom" as one word, and everything after the first # comment was treated as a comment and ignored. Because Part 1 failed, every later cell failed too, since they depended on variables such as X, y, and y_pred. I fixed it by putting each statement on its own line, then restarting the session and running all the cells in order. This taught me to read the error message and check formatting first, and to remember that notebook cells depend on each other.

Connection to My Goals

I am studying cybersecurity and plan to move toward AI security work, so this lab connected directly to where I want to go. In security, the cost of each kind of mistake is very different. If a malware detector or intrusion alert system produces a false negative, a real attack goes unnoticed, which can be very damaging. A false positive only means an analyst has to check an alert that turned out to be harmless. That makes recall the metric I would care about most in many security situations, though too many false positives can overwhelm a team and cause alert fatigue, so precision still matters. Learning to choose a metric based on the real cost of errors feels like a skill I will use in a security operations job.

Key Takeaways

My main takeaways are that accuracy by itself can be misleading, that the right metric depends on the cost of each type of error, and that cross-validation gives a more trustworthy estimate than a single split because it tests the model on several different parts of the data and averages the results. In the future, I will check the confusion matrix and cross-validate before trusting any model score.

Module 07 Lab - Model Evaluation Reflection Journal

Kei'Chara Fleeks-Jones

Knowledge Check Questions:

1. **Describe a real-world scenario where you would care more about a model's Precision than its Recall.**
    a. A spam filter for email is the example that I would use. If the filter has low precision, it will incorrectly label important emails as spam (False Positives), which can negatively impact the user. As someone who actively uses email filters to organize my emails and try to avoid spam as much as possible, I prefer to let a few spam emails slip into my inbox (tolerating lower recall) to guarantee that none of my important emails are falsely categorized as spam and permanently deleted.
2. **Describe a real-world scenario where you would care more about a model's Recall than its Precision.**
    a. A screening tool for a life-threatening illness in medical diagnostics. If the system misses a patient who has cancer, producing a False Negative, they may get overlooked and not receive any potentially life-saving treatment in a timely manner. So, we want to maximize recall to ensure we flag everyone who could potentially be sick. It's acceptable if a few healthy patients are falsely flagged, producing a False Positive, because follow up diagnostic tests and doctor visits should rule them out. The downside is potentially costing the patient more money.
3. **Why is a cross-validation score generally more trustworthy than a score from a single train-test split?**
    a. A single train-test split is susceptible to selection bias, otherwise known as luck. Depending on how the split is performed, the test set might randomly contain only easily predictable samples or

only hard to predict samples. Cross validation splits the data into multiple folds and trains/evaluates the model multiple times, ensuring every single data point gets to be in a test set exactly once.

In this lab I learned to go beyond simple accuracy to evaluate classification models more in depth. By adding a confusion matrix, I learned how to visually dissect true positives, true negatives, and the distinct real world impacts of Type I/False Positive and Type II/False Negative errors. Generating the classification report highlighted the trade offs between precision and recall. Finally, implementing 5-fold cross validation emphasized the importance of safeguarding against selection bias, showing that averaging performance across multiple dataset folds yields a far more trustworthy and stable indicator of a model's true generalization capabilities.